

Mode 3

$$\text{DP8 } (16 \times 8) \rightarrow \text{DP8 } (16 \times 4) \rightarrow \text{DP8 } (8 \times 4)$$

Step 1 — DP8 (16×8)

$$\text{DP8 } (16 \times 8) = \sum_{i=0}^7 a_i b_i$$

Step 2 — DP8 (16×4)

$$\text{DP8 } (16 \times 8) = 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 } (16 \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 } (16 \times 4)}$$

Step 3 — DP8 (8×4)

$$\text{DP8 } (16 \times 8) = 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hi}}_{\text{DP8 } (8 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lo}}_{\text{DP8 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lo}}_{\text{DP8 } (8 \times 4)}$$